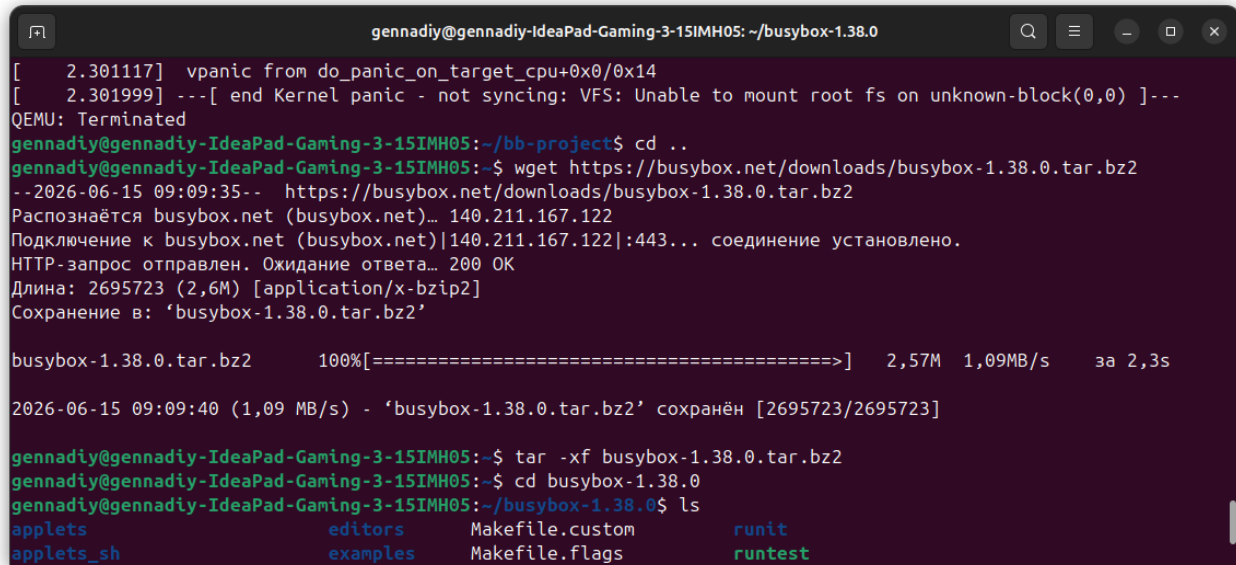


Задание 18 — Корневая файловая система (часть 2, init/BusyBox)

1) Загрузка BusyBox



```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/busybox-1.38.0
[ 2.301117] vpanic from do_panic_on_target_cpu+0x0/0x14
[ 2.301999] ---[ end Kernel panic - not syncing: VFS: Unable to mount root fs on unknown-block(0,0) ]---
QEMU: Terminated
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/bb-project$ cd ..
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$ wget https://busybox.net/downloads/busybox-1.38.0.tar.bz2
--2026-06-15 09:09:35-- https://busybox.net/downloads/busybox-1.38.0.tar.bz2
Распознаётся busybox.net (busybox.net)... 140.211.167.122
Подключение к busybox.net (busybox.net)|140.211.167.122|:443... соединение установлено.
HTTP-запрос отправлен. Ожидание ответа... 200 OK
Длина: 2695723 (2,6М) [application/x-bzip2]
Сохранение в: 'busybox-1.38.0.tar.bz2'

busybox-1.38.0.tar.bz2      100%[=====] 2,57M  1,09MB/s   за 2,3s
2026-06-15 09:09:40 (1,09 MB/s) - 'busybox-1.38.0.tar.bz2' сохранён [2695723/2695723]

gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$ tar -xvf busybox-1.38.0.tar.bz2
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$ cd busybox-1.38.0
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/busybox-1.38.0$ ls
applets          editors          Makefile.custom  runit
applets_sh       examples         Makefile.flags   runtest
```

Рисунок 1 — процесс загрузки исходного кода BusyBox

2) Создаем init.c и выполняем компиляция с ключом -static. Затем упаковываем во временную файловую систему initramfs. И запускаем ядро в эмуляторе QEMU с передачей initramfs.cpio.gz с флагом -initrd:



```
GNU nano 7.2                                init.c *
#include <stdio.h>
#include <unistd.h>

int main() {
    printf("HELL000!\n");
    while(1) {
        sleep(20);
    }
    return 0;
}
```

^G Справка ^O Записать ^W Поиск ^K Вырезать ^T Выполнить ^C Позиция M-U Отмена
^X Выход ^R ЧитФайл ^_ Замена ^U Вставить ^J Выводить ^/ К строке M-E Повтор

Рисунок 2 — исходный код init.c в gnu nano

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/bb-project
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/bb-project$ arm-linux-gnueabi-gcc -static init.c -o init
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/bb-project$ ls -la
итого 12176
drwxrwxr-x  2 gennadiy gennadiy   4096 июн 15 10:23 .
drwxr-x--- 24 gennadiy gennadiy   4096 июн 15 09:09 ..
-rwxrwxr-x  1 gennadiy gennadiy  424116 июн 15 10:23 init
-rw-rw-r--  1 gennadiy gennadiy    135 июн 15 10:22 init.c
-rw-rw-r--  1 gennadiy gennadiy   14329 июн 15 07:39 vexpress-v2p-ca9.dtb
-rwxrwxr-x  1 gennadiy gennadiy 12009984 июн 15 06:59 zImage
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/bb-project$ echo init | cpio -H newc -o | gzip > initramfs.cpio.gz
829 блоков
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/bb-project$ ls -la
итого 12400
drwxrwxr-x  2 gennadiy gennadiy   4096 июн 15 10:26 .
drwxr-x--- 24 gennadiy gennadiy   4096 июн 15 09:09 ..
-rwxrwxr-x  1 gennadiy gennadiy  424116 июн 15 10:23 init
-rw-rw-r--  1 gennadiy gennadiy    135 июн 15 10:22 init.c
-rw-rw-r--  1 gennadiy gennadiy  225514 июн 15 10:26 initramfs.cpio.gz
-rw-rw-r--  1 gennadiy gennadiy   14329 июн 15 07:39 vexpress-v2p-ca9.dtb
-rwxrwxr-x  1 gennadiy gennadiy 12009984 июн 15 06:59 zImage
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/bb-project$
```

Рисунок 3 — сборка временной файловой системы

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/bb-project
[ 2.206188] Key type dns_resolver registered
[ 2.206907] ThumbEE CPU extension supported.
[ 2.207019] Registering SWP/SWPB emulation handler
[ 2.247895] Loading compiled-in X.509 certificates
[ 2.278847] 10009000.serial: ttyAMA0 at MMIO 0x10009000 (irq = 37, base_baud = 0) is a PL011 rev 1
[ 2.283671] printk: console [ttyAMA0] enabled
[ 2.345916] 1000a000.serial: ttyAMA1 at MMIO 0x1000a000 (irq = 38, base_baud = 0) is a PL011 rev 1
[ 2.354396] 1000b000.serial: ttyAMA2 at MMIO 0x1000b000 (irq = 39, base_baud = 0) is a PL011 rev 1
[ 2.359388] 1000c000.serial: ttyAMA3 at MMIO 0x1000c000 (irq = 40, base_baud = 0) is a PL011 rev 1
[ 2.435354] input: ImExPS/2 Generic Explorer Mouse as /devices/platform/bus@40000000/bus@40000000:motherboard-bus@40000000/bus@40000000:motherboard-bus@40000000:iofpga@7,00000000/10007000.kmi/serio1/input/input1
[ 2.506213] clk: Disabling unused clocks
[ 2.506554] PM: genpd: Disabling unused power domains
[ 2.574444] Freeing unused kernel image (initmem) memory: 2048K
[ 2.577198] input: AT Raw Set 2 keyboard as /devices/platform/bus@40000000/bus@40000000:motherboard-bus@40000000/bus@40000000:motherboard-bus@40000000:iofpga@7,00000000/10006000.kmi/serio0/input/input2
[ 2.593604] Run /init as init process
HELLOOO!
[ 12.653171] amba 1000f000.wdt: deferred probe pending: (reason unknown)
[ 12.653252] amba 100e0000.memory-controller: deferred probe pending: (reason unknown)
[ 12.653266] amba 100e1000.memory-controller: deferred probe pending: (reason unknown)
[ 12.653277] amba 100e5000.watchdog: deferred probe pending: (reason unknown)
```

Рисунок 4 — выполнение пользовательского процесса init в эмуляторе QEMU

3) Выполняем команду `make ARCH=arm CROSS_COMPILE=arm-linux-gnueabi-gcc defconfig` для конфигурации исходного кода BusyBox 1.38.0

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/busybox-1.38.0
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/bb-project$ cd ..
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$ cd busybox-1.38.0/
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/busybox-1.38.0$ make ARCH=arm CROSS_COMPILE=arm-linux-gnueabi- defconfig
HOSTCC scripts/basic/fixdep
HOSTCC scripts/basic/split-include
HOSTCC scripts/basic/docproc
GEN include/applets.h
GEN include/usage.h
GEN scripts/Kbuild
GEN klibc-utils/Kbuild
GEN klibc-utils/Config.in
GEN shell/Kbuild
GEN shell/Config.in
GEN mailutils/Kbuild
GEN mailutils/Config.in
GEN applets/Kbuild
GEN miscutils/Kbuild
GEN miscutils/Config.in
GEN loginutils/Kbuild
GEN loginutils/Config.in
GEN syslogd/Kbuild
GEN syslogd/Config.in
GEN console-tools/Kbuild
GEN console-tools/Config.in
GEN e2fsprogs/Kbuild
GEN e2fsprogs/Config.in
GEN modutils/Kbuild
GEN modutils/Config.in
```

Рисунок 5 — подготовка конфигурационного файла

4) Выполняем команду `make ARCH=arm CROSS_COMPILE=arm-linux-gnueabi- menuconfig` для включения статической сборки

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/busybox-1.38.0
BusyBox 1.38.0 Configuration

Settings
Arrow keys navigate the menu. <Enter> selects submenus ---. Highlighted letters are
hotkeys. Pressing <Y> includes, <N> excludes, <M> modularizes features. Press
<Esc><Esc> to exit, <?> for Help, </> for Search. Legend: [*] built-in [ ] excluded
<M> module < > module capable

[*] Drop SUID state for most applets
[*] Enable SUID configuration via /etc/busybox.conf
[*] Suppress warning message if /etc/busybox.conf is not readable
[ ] exec prefers applets
(/proc/self/exe) Path to busybox executable
[ ] Support NSA Security Enhanced Linux
[ ] Clean up all memory before exiting (usually not needed)
[*] Support LOG_INFO level syslog messages
--- Build Options
[*] Build static binary (no shared libs)
[ ] Force NOMMU build
(arm-linux-gnueabi-) Cross compiler prefix
() Path to sysroot
() Additional CFLAGS
() Additional LDFLAGS

<Select> < Exit > < Help >
```

Рисунок 6 — настройка конфигурации

5) Запускаем процесс компиляции и установки встроенных утилит командой `make install -j8`

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/busybox-1.38.0

*** End of configuration.
*** Execute 'make' to build the project or try 'make help'.

gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/busybox-1.38.0$ make install -j8
SPLIT  include/autoconf.h -> include/config/*
GEN     include/bbconfigopts.h
GEN     include/common_bufsiz.h
GEN     include/embedded_scripts.h
HOSTCC  applets/usage
HOSTCC  applets/applet_tables
applets/usage.c: In function 'main':
applets/usage.c:52:17: warning: ignoring return value of 'write' declared with attribute 'warn_unused_result' [-Wunused-result]
   52 |                 write(STDOUT_FILENO, usage_array[i].usage, strlen(usage_array[i].usage) + 1
      |                 ^
      |
GEN     include/usage_compressed.h
GEN     include/applet_tables.h include/NUM_APPLETS.h
GEN     include/applet_tables.h include/NUM_APPLETS.h
HOSTCC  applets/usage_pod
CC       applets/applets.o
LD       applets/built-in.o
LD       archival/built-in.o
CC       archival/bbunzip.o
CC       archival/bzip2.o
CC       archival/chksum_and_xwrite_tar_header.o
CC       archival/cpio.o
CC       archival/cpio.o
```

Рисунок 7 — процесс компиляции и установки утилит BusyBox

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/busybox-1.38.0

./_install//usr/sbin/remove-shell -> ../../bin/busybox
./_install//usr/sbin/rtctwake -> ../../bin/busybox
./_install//usr/sbin/secdrng -> ../../bin/busybox
./_install//usr/sbin/sendmail -> ../../bin/busybox
./_install//usr/sbin/setfont -> ../../bin/busybox
./_install//usr/sbin/setlogcons -> ../../bin/busybox
./_install//usr/sbin/svlogd -> ../../bin/busybox
./_install//usr/sbin/telnetd -> ../../bin/busybox
./_install//usr/sbin/tftpd -> ../../bin/busybox
./_install//usr/sbin/ubiattach -> ../../bin/busybox
./_install//usr/sbin/ubidetach -> ../../bin/busybox
./_install//usr/sbin/ubimkvol -> ../../bin/busybox
./_install//usr/sbin/ubirename -> ../../bin/busybox
./_install//usr/sbin/ubirmvol -> ../../bin/busybox
./_install//usr/sbin/ubirsvol -> ../../bin/busybox
./_install//usr/sbin/ubiupdatevol -> ../../bin/busybox
./_install//usr/sbin/udhcpd -> ../../bin/busybox

-----
You will probably need to make your busybox binary
setuid root to ensure all configured applets will
work properly.
-----

gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/busybox-1.38.0$ file busybox
busybox: ELF 32-bit LSB executable, ARM, EABI5 version 1 (GNU/Linux), statically linked, BuildID[sha1]=562793c2ec16f4f3e3f3a759d47ad63f96adfb36, for GNU/Linux 3.2.0, stripped
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/busybox-1.38.0$
```

Рисунок 7 — результат компиляции и установки


```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/busybox-1.38.0/_install

-----
You will probably need to make your busybox binary
setuid root to ensure all configured applets will
work properly.
-----

gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/busybox-1.38.0$ file busybox
busybox: ELF 32-bit LSB executable, ARM, EABI5 version 1 (GNU/Linux), statically linked, BuildID[sha1]=562793c2ec16f4f3e3f3a759d47ad63f96adfb36, for GNU/Linux 3.2.0, stripped
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/busybox-1.38.0$ ls
applets          coreutils        LICENSE          procps
applets_sh       debianutils      loginutils      qemu_multiarch_testing
arch             docs             mailutils       README
archival         e2fsprogs        Makefile        runit
AUTHORS          editors          Makefile.custom runtest
busybox          examples         Makefile.flags  scripts
busybox_ldscript.README.txt findutils        Makefile.help   selinux
busybox.links    include          make_single_applets.sh shell
busybox_unstripped init             miscutils       size_single_applets.sh
busybox_unstripped.map _install        modutils        sysklogd
busybox_unstripped.out INSTALL          networking      testsuite
Config.in        klibc-utils     NOFORK_NOEXEC.lst TODO
configs          libbb           NOFORK_NOEXEC.sh TODO_unicode
console-tools    libpwdgrp       printutils      util-linux

gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/busybox-1.38.0$ cd _install/
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/busybox-1.38.0/_install$ ls
bin linuxrc sbin usr
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/busybox-1.38.0/_install$
```

Рисунок 8 — результат компиляции и установки

6) Полученную структуру корневой файловой системы в _install архивируем и копируем в рабочую директорию bb-project. Далее выполняем запуск системы в эмуляторе командой QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kernel zImage -dtb vexpress-v2p-ca9.dtb -initrd initramfs.cpio.gz -append "console=ttyAMA0 rdinit=/bin/ash" -nographic

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/bb-project

gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/bb-project$ cd ..
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$ cd busybox-1.38.0/
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/busybox-1.38.0$ cd _install/
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/busybox-1.38.0/_install$ find . | cpio -H newc -o | g
zip > initramfs.cpio.gz
3159 блоков
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/busybox-1.38.0/_install$ ls
bin initramfs.cpio.gz linuxrc sbin usr
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/busybox-1.38.0/_install$ cp initramfs.cpio.gz ~/bb-pro
ject/
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/busybox-1.38.0/_install$ ls
bin initramfs.cpio.gz linuxrc sbin usr
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/busybox-1.38.0/_install$ cd ..
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/busybox-1.38.0$ cd ..
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$ cd bb-project/
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/bb-project$ QEMU_AUDIO_DRV=none qemu-system-arm -M vex
press-a9 -kernel zImage -dtb vexpress-v2p-ca9.dtb -initrd initramfs.cpio.gz -append "console=ttyAMA
0 rdinit=/bin/ash" -nographic
[ 0.000000] Booting Linux on physical CPU 0x0
[ 0.000000] Linux version 7.0.12 (gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05) (arm-linux-gnueabi
hf-gcc (Ubuntu 13.3.0-6ubuntu2-24.04.1) 13.3.0, GNU ld (GNU Binutils for Ubuntu) 2.42) #1 SMP Mon J
un 15 06:58:46 +07 2026
[ 0.000000] CPU: ARMv7 Processor [41050003] revision 0 (ARMv7), 400 MHz
[ 0.000000] CPU: 0x41050003
```

Рисунок 9 — сборка архива initramfs на базе BusyBox и его запуск в эмуляторе QEMU

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/bb-project
0:motherboard-bus@40000000/bus@40000000:motherboard-bus@40000000:iofpga@7,00000000/10007000.kmi/ser
io1/input/input2
/bin/ash: can't access tty; job control turned off
~ # [ 12.630875] amba 100f000.wdt: deferred probe pending: (reason unknown)
[ 12.631128] amba 100e0000.memory-controller: deferred probe pending: (reason unknown)
[ 12.631180] amba 100e1000.memory-controller: deferred probe pending: (reason unknown)
[ 12.631216] amba 100e5000.watchdog: deferred probe pending: (reason unknown)

~ # ls
bin          initramfs.cpio.gz  root          usr
dev          linuxrc           sbin
~ # ls -l
total 0
drwxrwxr-x  2 1000    1000          1920 Jun 15 04:48 bin
drwxr-xr-x  2 0        0              60 Jun 14 23:49 dev
-rw-rw-r--  1 1000    1000              0 Jun 15 05:49 initramfs.cpio.gz
lrwxrwxrwx  1 1000    1000              11 Jun 15 04:48 linuxrc -> bin/busybox
drwx-----  2 0        0              40 Jun 14 23:49 root
drwxrwxr-x  2 1000    1000         1460 Jun 15 04:48 sbin
drwxrwxr-x  4 1000    1000              80 Jun 15 04:48 usr
~ #
~ #
~ #
```

Рисунок 10 — результаты выполнения (проверка командой ls)

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/bb-project
total 0
drwxrwxr-x  2 1000    1000          1920 Jun 15 04:48 bin
drwxr-xr-x  2 0        0              60 Jun 14 23:49 dev
-rw-rw-r--  1 1000    1000              0 Jun 15 05:49 initramfs.cpio.gz
lrwxrwxrwx  1 1000    1000              11 Jun 15 04:48 linuxrc -> bin/busybox
drwx-----  2 0        0              40 Jun 14 23:49 root
drwxrwxr-x  2 1000    1000         1460 Jun 15 04:48 sbin
drwxrwxr-x  4 1000    1000              80 Jun 15 04:48 usr
~ #
~ #
~ # mkdir proc
~ # mount -t proc proc proc
~ # ps
PID  USER    TIME  COMMAND
  1  0        0:00  /bin/ash
  2  0        0:00  [kthreadd]
  3  0        0:00  [pool_workqueue_]
  4  0        0:00  [kworker/R-rcu_g]
  5  0        0:00  [kworker/R-sync_]
  6  0        0:00  [kworker/R-kvfre]
  7  0        0:00  [kworker/R-slub_]
  8  0        0:00  [kworker/R-netns]
 10  0        0:01  [kworker/0:1-eve]
```

Рисунок 11 — результаты выполнения (примонтировали виртуальную файловую систему proc)